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# Markdown Studio — Feature Demo

Open this file and run **Markdown Studio: Preview** ( `Cmd+Shift+P` ).

---

## 1. Markdown Rendering

**Bold**, *italic*, ~~strikethrough~~, `inline code`

> Blockquotes work too.

| Feature | Status |
|---|---|
| Markdown | ✅ |
| Mermaid | ✅ |
| PlantUML | ✅ |
| SVG | ✅ |
| PDF Export | ✅ |

1. Ordered list item
2. Another item
3. Third item

- Unordered item
- Another item

- Item

### 日本語テスト

- 日本語あ `いう` えお
  - 2byte文字 `テスト`

### Task Lists

- [x] Markdown rendering
- [x] Mermaid diagrams
- [x] PlantUML diagrams
- [x] LaTeX math
- [x] Footnotes
- [ ] no checked

## Footnotes

This sentence has a footnote[1] and another one[2].

## Emoji

😄 🚀 👍 ⚠️ ❤️ ⭐ 🔥 ☕

## Math (KaTeX)

Inline math: $E = mc^2$ and $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

Display math:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

## Definition Lists

**Markdown**
: A lightweight markup language for creating formatted text.

**PlantUML**
: A tool for creating UML diagrams from plain text.

  Uses Smetana layout engine (no Graphviz needed).

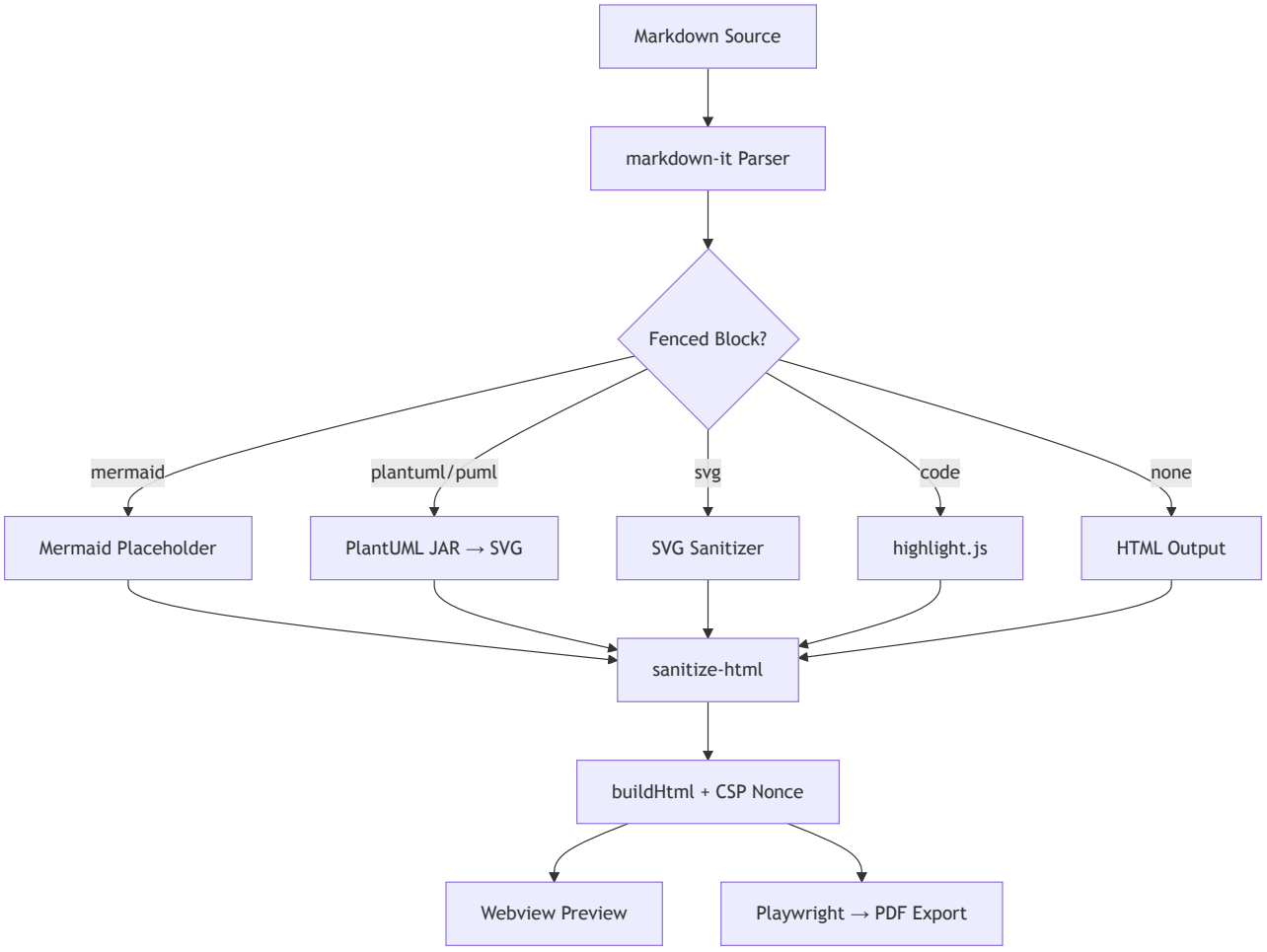
## Superscript / Subscript

- Water: $H_2O$
- Einstein: $E = mc^2$
- 19th century

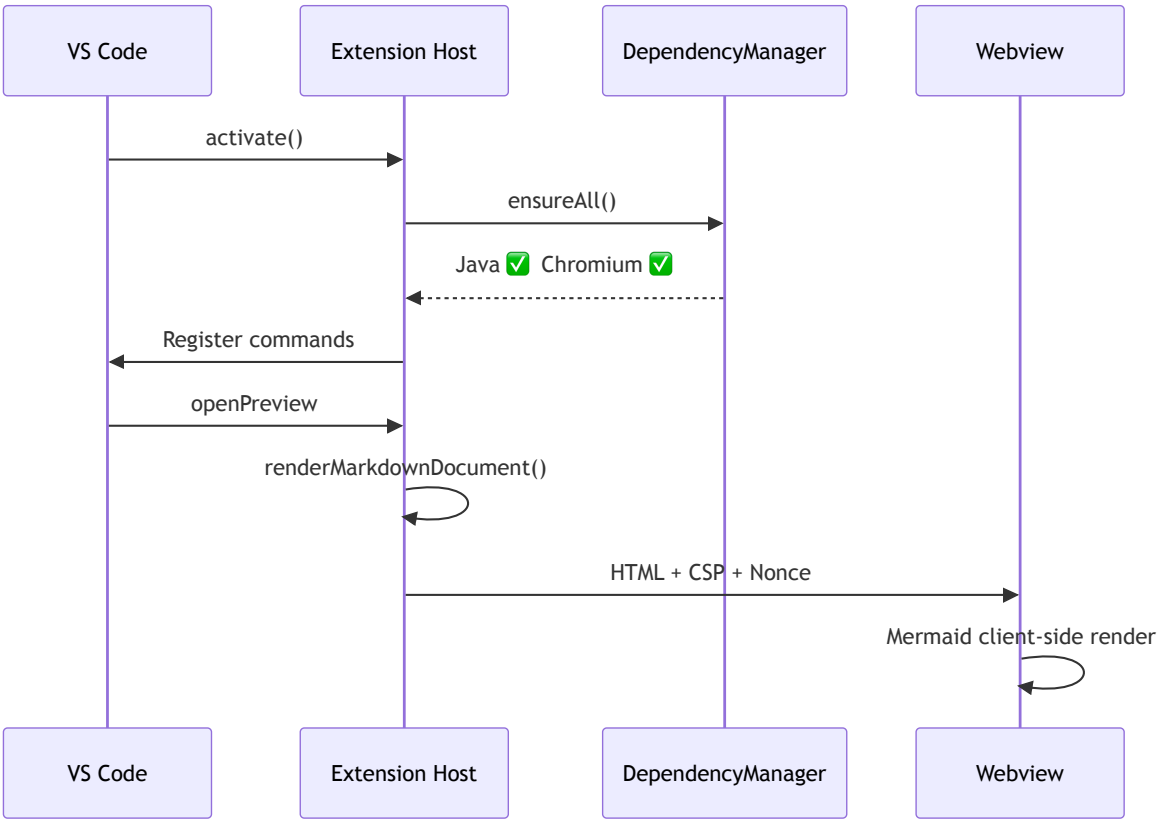
---

# 2. Mermaid Diagrams

## Markdown Studio Architecture



### Extension Activation Flow

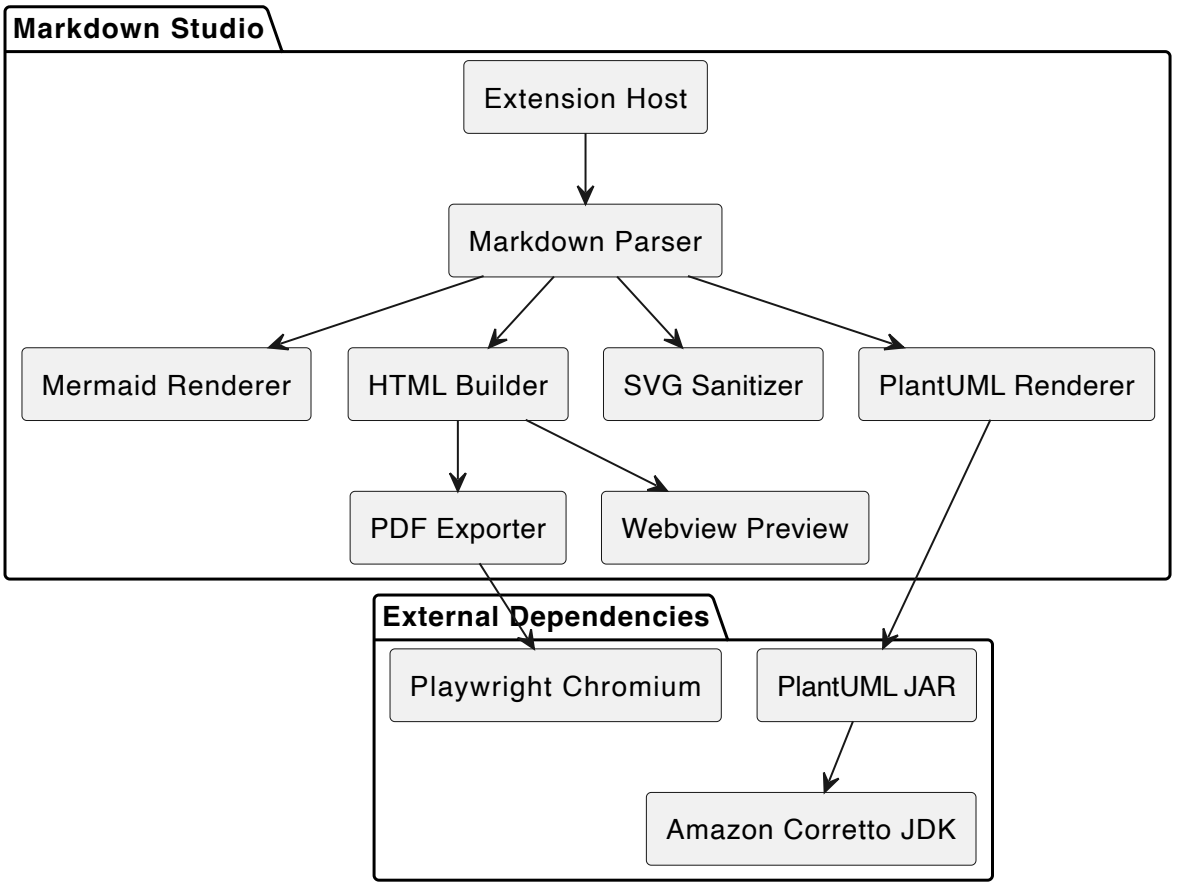



---

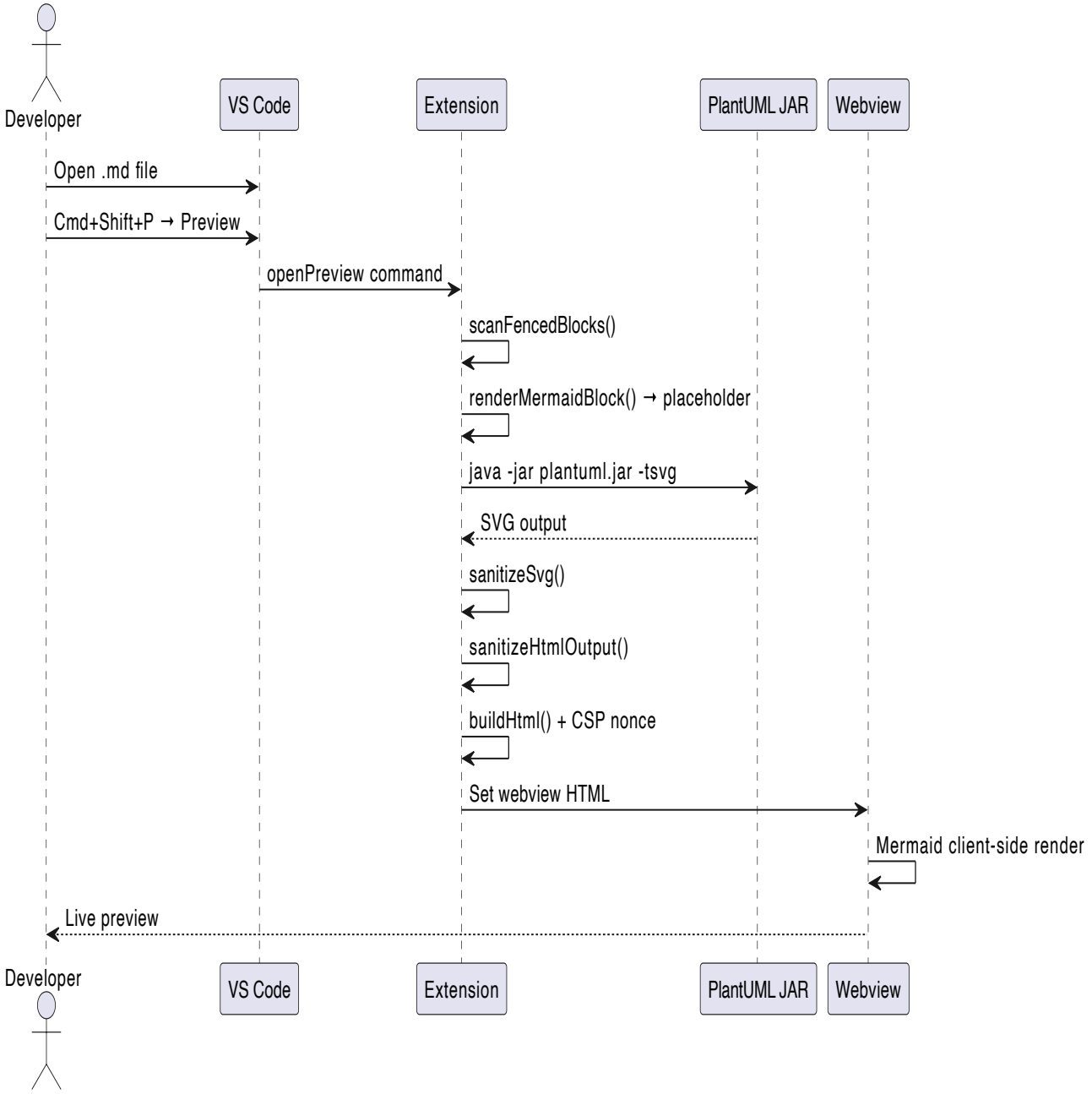
## 3. PlantUML Diagrams

Rendered locally via bundled JAR. No remote server.

## Extension Component Diagram



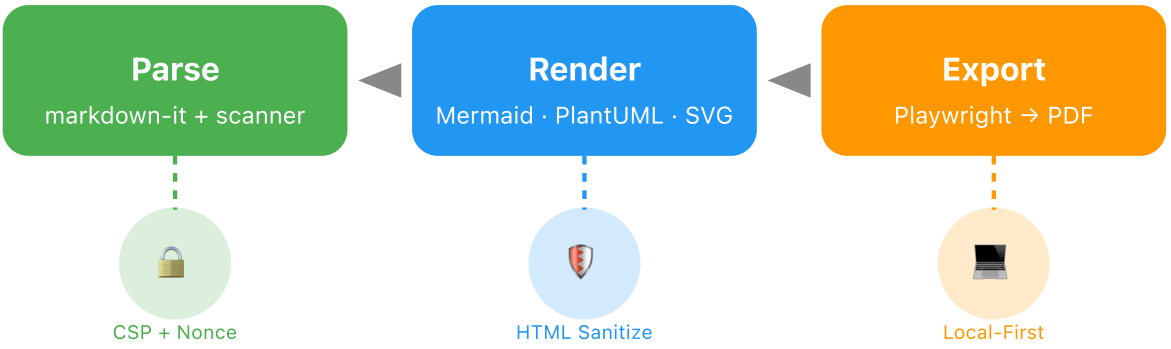
### Document Processing Sequence




---

## 4. Inline SVG (Sanitized)

Dangerous elements ( `<script>` , `<foreignObject>` , event handlers) are stripped automatically.




---

## 5. Syntax Highlighting

```
import * as vscode from 'vscode';

export function activate(context: vscode.ExtensionContext): void {
  const depManager = new DependencyManager();
  const status = await depManager.ensureAll(context);

  context.subscriptions.push(
    vscode.commands.registerCommand('markdownStudio.openPreview', async () => {
      await openPreviewCommand(context);
    }),
    vscode.commands.registerCommand('markdownStudio.exportPdf', async () => {
      await exportPdfCommand(context);
    })
  );
}
```

```
{
  "markdownStudio.plantuml.mode": "bundled-jar",
  "markdownStudio.java.path": "java",
  "markdownStudio.export.pageFormat": "A4",
  "markdownStudio.security.blockExternalLinks": true
}
```

```
# PlantUML rendering is also useful for Python projects
from dataclasses import dataclass

@dataclass
class DiagramConfig:
    mode: str = "bundled-jar"
    java_path: str = "java"
    timeout_ms: int = 15000
```

```
fmt.Println("Hello, World!")
```

```
plain text without language
```

```rust
fn main() {
    println!("Hello from Rust");
}
```

```sql
SELECT id, name FROM users WHERE active = true;
```

```dockerfile
FROM node:20-alpine
WORKDIR /app
COPY . .
RUN npm install
```

---

## 6. Images and Security

Markdown Studio is local-first. The default security policy ( `blockExternalLinks = true` ) blocks external resources while allowing local files.

### ✅ Local image (relative path — works)

- local svg

- icon.png

- icon.svg

Relative paths from the Markdown file are resolved and displayed correctly.

## ✅ Local image (absolute path — works)

Images with absolute file paths also work in the preview.

## ❌ External image (blocked by policy)

External image blocked by policy.

External URLs ( `https://...` ) are blocked by default. The image above shows a policy notice instead.

## ❌ External link (blocked by policy)

External link — blocked by default 🖐

External links are replaced with a blocked notice in the preview.

### How to allow external resources

Set `markdownStudio.security.blockExternalLinks` to `false` in VS Code settings to allow external images and links. This is not recommended for corporate/secure environments.

### Security Summary

| Resource Type | Default | Configurable |
|---|---|---|
| Local images (relative path) | ✅ Allowed | — |
| Local images (absolute path) | ✅ Allowed | — |
| External images (https://) | ❌ Blocked | `blockExternalLinks: false` |
| External links (https://) | ❌ Blocked | `blockExternalLinks: false` |
| Inline SVG | ✅ Allowed | — |
| Mermaid diagrams | ✅ Allowed | — |
| PlantUML diagrams | ✅ Allowed | — |

## 7. Theme Adaptability

Switch between **light** and **dark** mode in VS Code ( `Cmd+K Cmd+T` ) to see how the preview adapts:

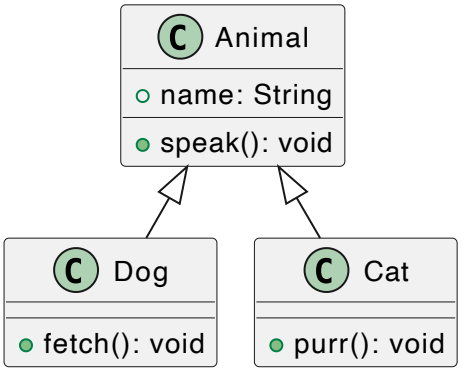
- Mermaid diagrams automatically switch between light and dark themes
- SVG elements use colors chosen for visibility in both themes
- Code blocks use theme-aware syntax highlighting
- PlantUML output receives CSS overrides for dark mode

---

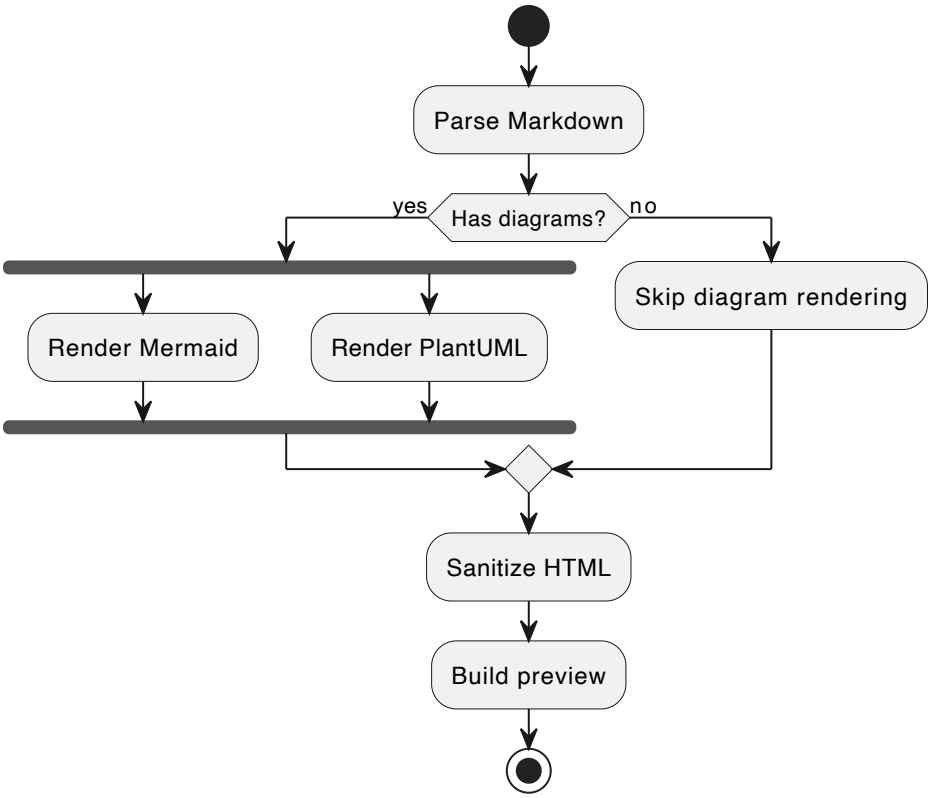
## 8. Diagram Type Catalog

All diagram types below are verified to render correctly with the bundled PlantUML + Smetana engine (no Graphviz required).

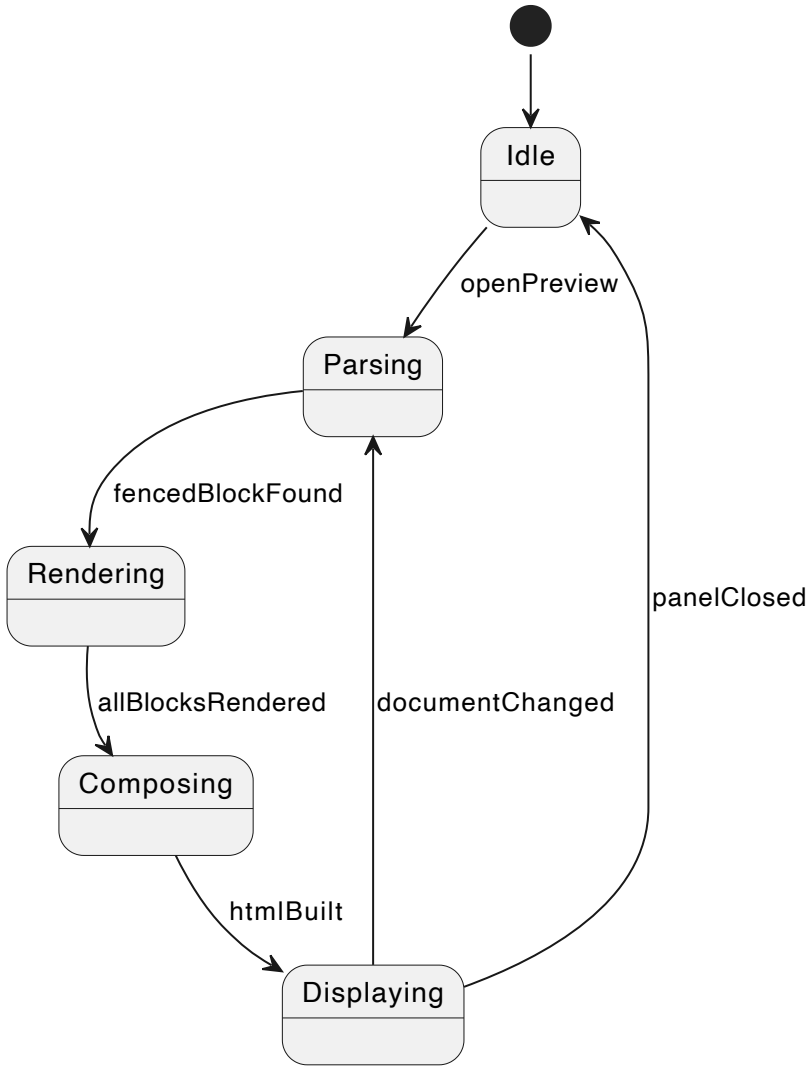
▼ Class Diagram



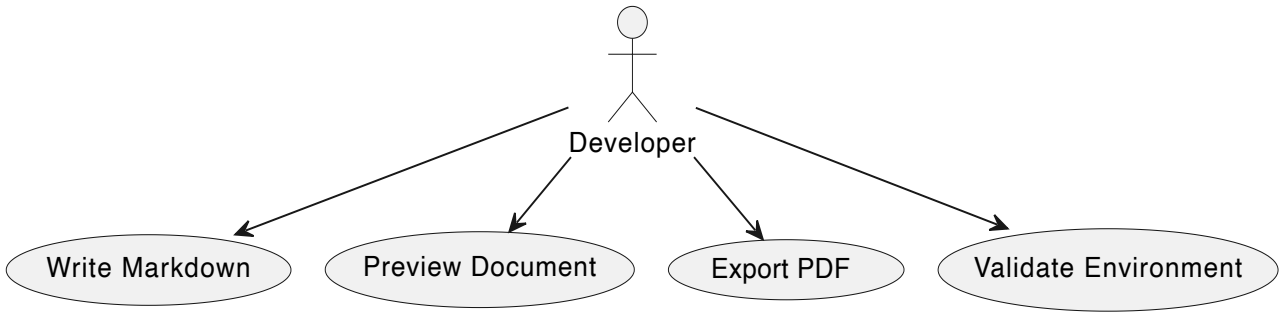

▼ Activity Diagram



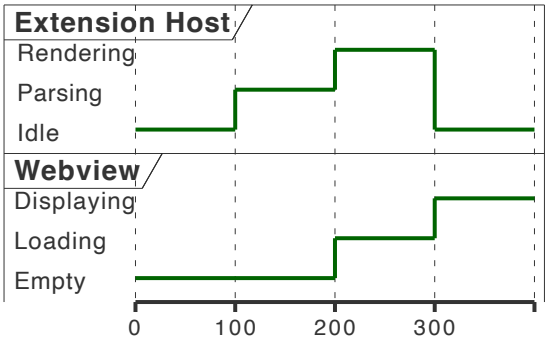

▼ State Diagram



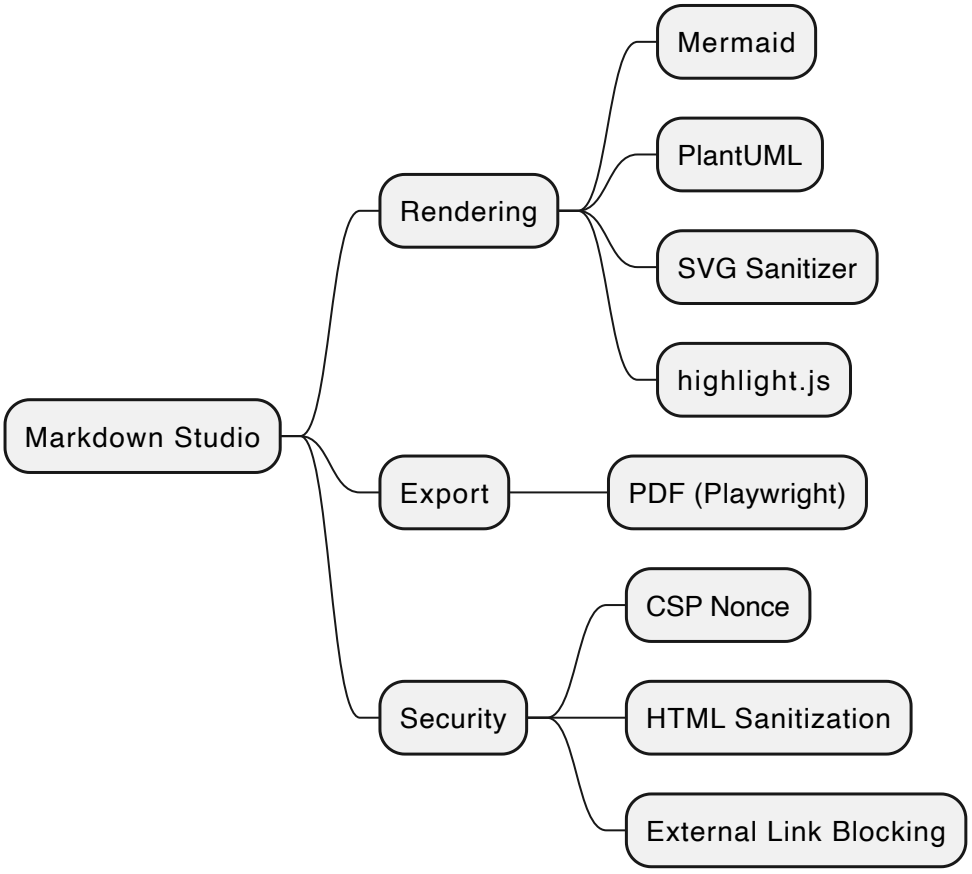

▼ Use Case Diagram



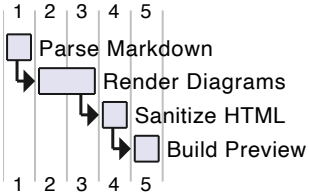

▼ Timing Diagram



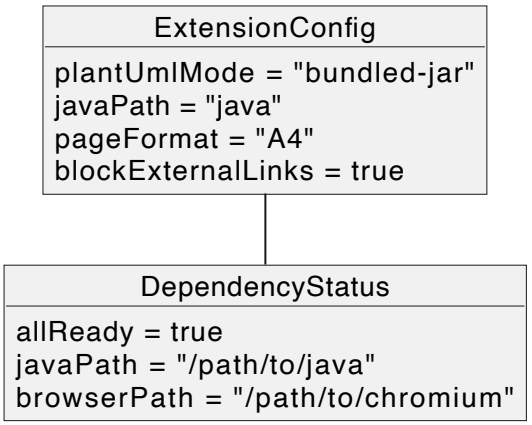

▼ Mind Map



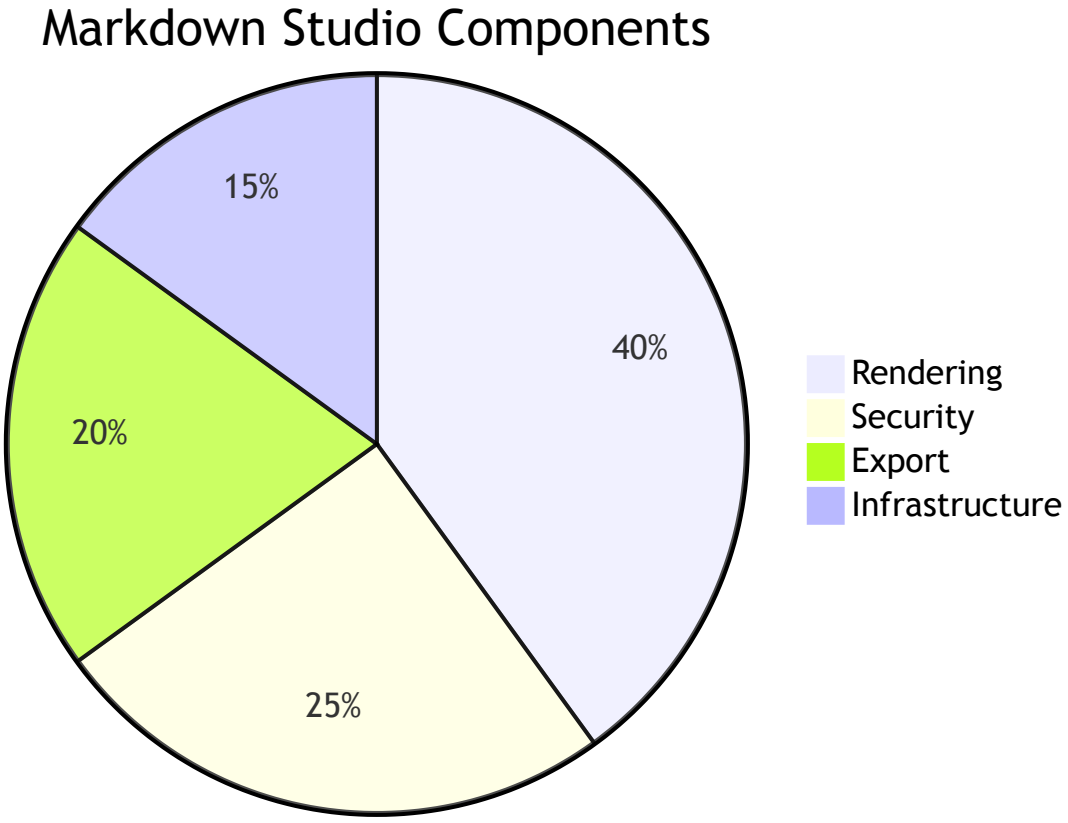

▼ Gantt Chart



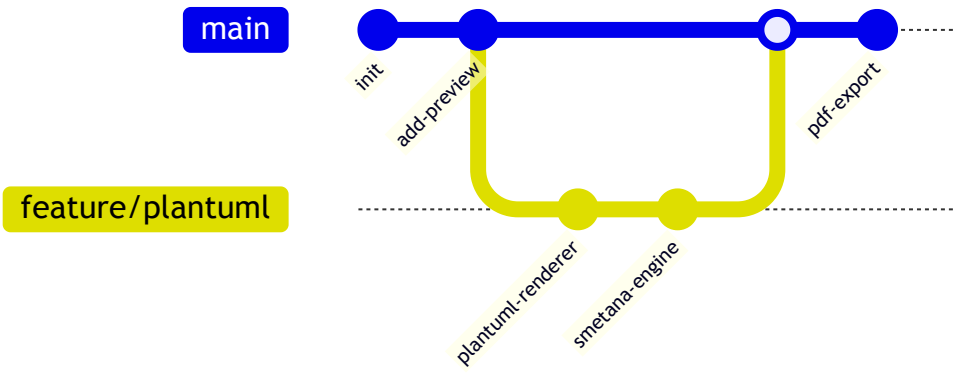

▼ Object Diagram



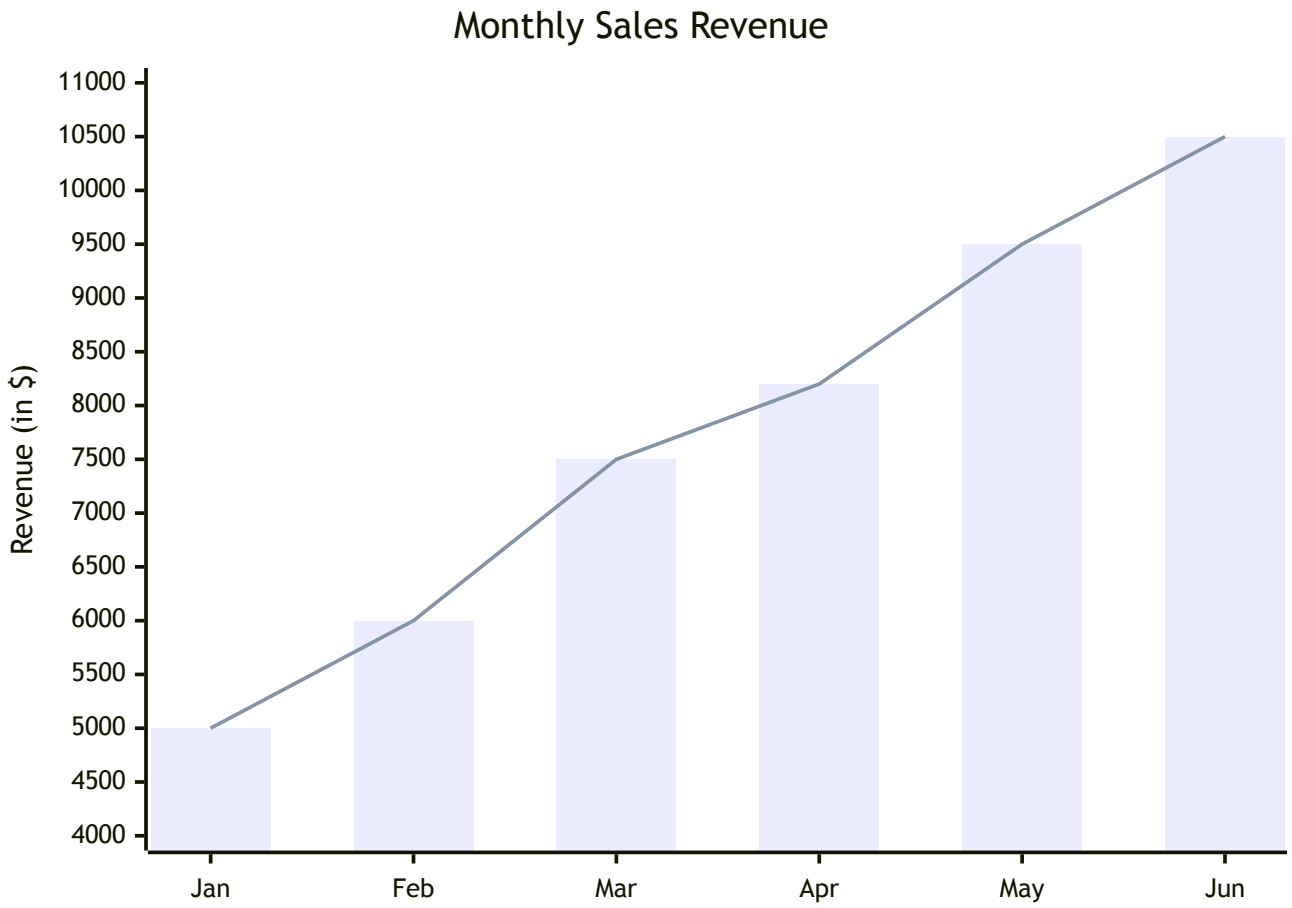

▼ Mermaid: Pie Chart



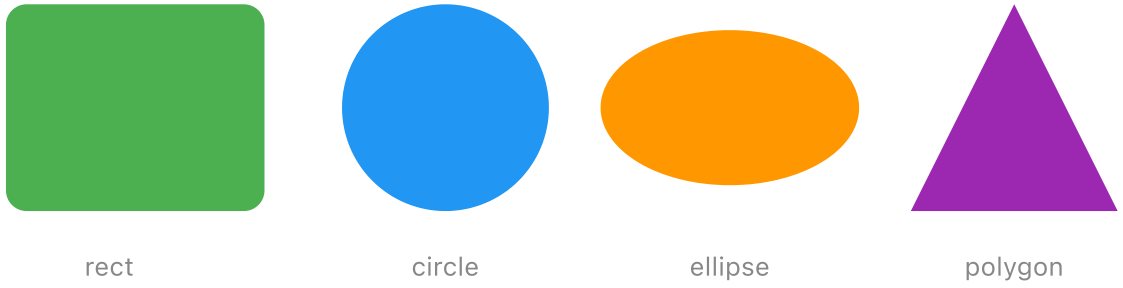

▼ Mermaid: Git Graph




▼ Mermaid: XY Chart

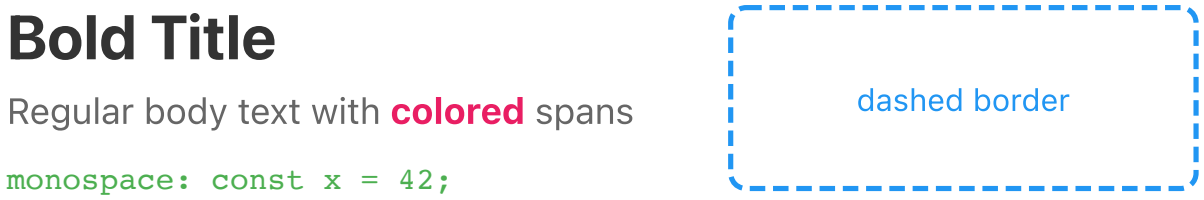



▼ SVG: Basic Shapes

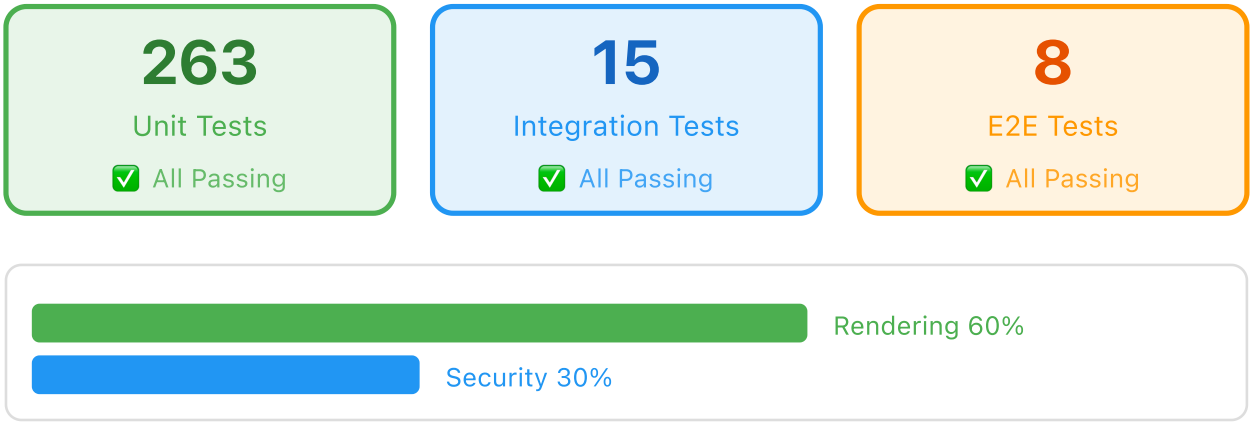



▼ SVG: Lines and Paths

▼ SVG: Text and Styling




▼ SVG: Dashboard Layout




---

## 9. Custom CSS

Markdown Studio supports built-in themes and inline CSS customization.

### Bundled Themes

Select a theme from the settings dropdown ( `markdownStudio.style.theme` ):

| Theme | Description |
|---|---|
| `default` | No extra styling — uses the selected preset only |
| `modern` | Indigo accents, soft shadows, refined typography |
| `markdown-pdf` | Classic look matching the popular Markdown PDF extension |
| `minimal` | Bare-bones, clean starting point |

All themes include dark mode support and print-optimized styles.

### Setup

```
// Pick a theme from the dropdown
"markdownStudio.style.theme": "modern"

// Optionally add your own CSS on top
"markdownStudio.style.customCss": "h1 { color: navy; } blockquote { border-left: 3px solid orange; }"
```

### How It Works

- Theme CSS is injected **after** the preset styles, so theme rules take priority
- `customCss` is applied **after** the theme, so your inline rules override everything
- The same CSS applies to both preview and PDF export for consistent output
- Copy any bundled theme and customize it to create your own

---

## 10. PDF Export

This entire document can be exported to PDF:

1. Open this file in VS Code
2. `Cmd+Shift+P` → **Markdown Studio: Export PDF**
3. A `demo.pdf` will be generated next to this file

The PDF uses the same HTML pipeline as the preview.

---

*Markdown Studio v0.7.0*

---

1. This is the first footnote content. ↩

2. This is the second footnote with **bold** text. ↩